

Towards Integrated AI-Assisted Legal Reasoning: A Survey of Indian Legal NLP Across Five Capability Axes

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Abstract—India’s courts are buried under nearly fifty million pending cases. The scale alone makes a strong case for computational intervention, yet the legal AI tools available to Indian practitioners today are built in isolation from one another: retrieval engines that know nothing about prediction models, prediction models that operate independently of explainability modules, and explainability components that rarely feed their signals back into retrieval. In this work, we survey Indian legal NLP through five capability axes, namely *precedent search*, *case fact structuring*, *argument graph construction*, *judicial outcome simulation*, and *counterfactual what-if analysis*. For each axis we catalogue representative methods, benchmark datasets, and evaluation practices. Our principal observation is blunt: no existing system unifies all five axes under a single coherent architecture, and at least two of them, argument graph construction and counterfactual reasoning, are for practical purposes absent from the Indian legal AI landscape. Whether this fragmentation is best read as an artefact of how the field grew up, axis by axis, or as a more fundamental modelling oversight, we will not pretend to have fully settled. What we do argue is that further progress on individual axes appears to have reached a point of diminishing returns, and that meaningful advancement now depends on a unifying pipeline that couples retrieval, structured reasoning, prediction, and interactive analysis.

I. INTRODUCTION

India’s courts are drowning. Roughly fifty million cases sit pending across district courts, high courts, and the Supreme Court of India, and the backlog grows each year faster than the judiciary can clear it. This is not, we believe, a scheduling problem; it is a structural failure of access to justice that periodic administrative reform has been unable to fix. Litigants wait years, sometimes a full decade, for resolution, and public trust erodes quietly, case by case. In this setting, AI-assisted legal tools offer a way to augment, not replace, judicial reasoning at a scale that human capacity alone cannot match. Over one million Supreme Court and High Court judgments are now digitised [1], providing an empirical base that few other common-law jurisdictions can rival. We believe these conditions make India one of the most consequential, and still underexplored, testbeds for legal AI research today.

The tools that do exist, however, talk past one another. A retrieval engine that surfaces relevant precedents has no awareness of the prediction model sitting one API call away. The prediction model, in turn, generates a disposition label without consulting how the retrieval module ranked its candidates. Explainability components, where they exist at all, are bolted

on after the fact. This fragmentation, we suspect, does not match how judicial deliberation actually unfolds in practice — though we should be honest that the line between how judges reason and how judges describe their reasoning in written opinions is one we have not fully resolved, and the literature we draw on here largely reads the latter as evidence for the former. A judge does not first retrieve, then apply statute, then weigh facts; she does all three at once, circling back the moment a new precedent reshapes the factual picture. Recent empirical evidence aligns with this intuition: coupling retrieval with downstream reasoning outperforms strict pipeline decomposition [2], and case-based reasoning integrated into retrieval-augmented generation produces answers that are measurably more grounded in legal sources [3]. The lesson, we take it, is that tight integration across capability boundaries is not optional but constitutive of legally faithful reasoning.

We should be upfront about what this paper is. It is a survey, not a system. Our contributions are twofold. First, we offer a structured review of five capability areas in Indian legal NLP, with attention to the methods, datasets, and evaluation practices that have shaped each. Second, we identify the missing integrated multi-capability pipeline as the field’s central open problem, and we sketch what such a pipeline would need to do. The five axes are: (1) *Precedent Search*, encompassing sparse and dense retrieval of prior judgments; (2) *Case Fact Structuring*, including named entity recognition, rhetorical role labelling, and statutory section tagging; (3) *Argument Graph Construction*, covering the extraction of claims, evidence, and their relational structure; (4) *Judicial Outcome Simulation*, predicting case dispositions alongside natural-language explanations; and (5) *What-If Analysis*, enabling counterfactual hypothesis testing over case facts. Taken together, these axes span the full reasoning chain from information retrieval to interactive deliberation.

The remainder of this paper follows a straightforward path. Section II sets up background on the Indian legal ecosystem and the specific difficulties it creates for NLP. Sections III through VII work through each capability axis in turn. Section VIII steps back to compare across axes, flagging integration gaps and recurring themes. Section IX takes up evaluation methodologies. Section X outlines open problems and future directions. Section XI concludes.

TABLE I
FIVE CAPABILITY AXES IN INDIAN LEGAL AI: COVERAGE AND IDENTIFIED GAPS

Capability Axis	Description	Gap in Existing Literature
Precedent Search	Semantic retrieval of relevant prior judgments through sparse keyword-based methods, dense vector representations, and hybrid strategies applied to Indian case law corpora.	Retrieval systems function as standalone components; their outputs are not consumed by downstream prediction or reasoning modules, limiting their utility in end-to-end legal analysis.
Case Fact Structuring	Extraction and classification of structured elements from judgment text, including named entity recognition, rhetorical role labelling of sentence-level functional segments, and Indian Penal Code section tagging.	Structured representations produced by these models are not propagated to any subsequent reasoning stage, rendering them terminal outputs rather than intermediate features in a larger pipeline.
Argument Graph Construction	Identification and relational mapping of argumentative components, including claims, supporting evidence, cited statutes, and invoked precedents, into directed graph representations that capture the logical structure of judicial reasoning.	This capability is virtually absent in Indian legal AI research; existing work on legal argument mining is confined almost exclusively to Western jurisdictions such as the European Court of Human Rights.
Judicial Outcome Simulation	Prediction of case dispositions (acceptance or dismissal) accompanied by natural language explanations that articulate the reasoning underlying the predicted outcome.	Prediction models operate without access to retrieval-augmented context; generated explanations remain shallow and lack grounding in specific precedents or statutory provisions.
What-If Analysis	Counterfactual hypothesis testing that enables users to perturb specific case facts, statutory provisions, or precedent selections and observe the resulting shifts in predicted outcomes and explanatory rationales.	Entirely unexplored in Indian legal NLP; no existing dataset, model, or evaluation framework addresses counterfactual reasoning over Indian case law.

II. BACKGROUND: THE INDIAN LEGAL ECOSYSTEM

India inherited its common law tradition from British colonial rule, and that inheritance carries real weight in everyday practice: judicial decisions are precedent, not merely advisory. Judicial reasoning here rests on three pillars that a court must hold in mind at once, namely the established facts of the case, the applicable statutory provisions (principally the Indian Penal Code and the Code of Civil Procedure), and binding prior rulings invoked through the doctrine of *stare decisis*. No competent judge consults these sequentially. A judgment forms at the intersection of all three: facts are read through the lens of statute, statutes are interpreted in light of prior rulings, and prior rulings are tested against the factual matrix of the case at hand. We contend that any computational framework which treats these pillars as independent inputs to a staged pipeline has, almost by construction, already compromised its fidelity to how law actually operates.

Processing Indian legal text introduces a set of difficulties that researchers working on ECHR or U.S. federal case law rarely encounter. The most immediate is multilingualism. A single judgment may shift between English, Hindi, Urdu, and a regional language mid-paragraph, with no markup or translation cues, which is a serious obstacle for tokenisers trained on monolingual corpora. Formatting is another persistent pain point. There is no standard template across Indian courts; judgments from different benches, different decades, and different levels of the judicial hierarchy vary substantially in structure, citation conventions, and rhetorical organisation, and rule-based extraction tends to break almost immediately under this variation. A third difficulty is annotation scarcity. The Legal-Eval shared task made this concrete: even trained annotators disagreed substantially on rhetorical role labels for Indian legal

text [4]. General-purpose encoders such as base BERT are not adequate to the task either. InLegalBERT demonstrated that domain-adapted pre-training on Indian corpora is essentially non-negotiable for competitive performance [5]. We suspect the community still underestimates how much domain shift matters in this setting — or, more honestly, perhaps the better way to put it is that domain shift is widely acknowledged in the abstract but consistently underfunded in practice, and we are not yet sure which framing leads to the right intervention.

The good news is that resource creation has accelerated sharply over the past five years. ILDC offered the community its first large-scale benchmark, providing over thirty-five thousand Supreme Court cases annotated for judgment prediction and explanation [6]. PredEx raised the bar by attaching expert-annotated explanations to specific textual spans within judgments [7]. NyayaAnumana, the most recent and largest corpus to date, spans multiple court levels and ships with its own domain-specialised language model [1]. Together, these datasets have driven measurable gains on tasks such as binary judgment prediction and rhetorical role classification. However, every one of them targets a single capability axis in isolation. No existing work unifies retrieval, structuring, reasoning, prediction, and interactive analysis under one coherent architecture. That gap, the absence of integration rather than the absence of data, is what motivates the five-axis taxonomy we develop in the sections that follow.

III. PRECEDENT SEARCH

Everything downstream depends on retrieval. In common law systems, prior rulings are not merely informative; they are binding. Finding the right precedent is the first move a lawyer makes when building an argument, and it is the step from which every other task in the pipeline inherits its quality, or by

which it is quietly poisoned. Grounding generation in retrieved case law produces markedly better outputs than letting a language model rely on its parametric memory alone [2], and pairing retrieval with case-based reasoning pushes legal fidelity further still [3]. We take it as fair to say that retrieval is the single highest-leverage point in any legal AI system: get it wrong, and nothing else in the pipeline matters.

A. Keyword-Based and Sparse Retrieval

For most Indian lawyers, precedent discovery still means typing terms into Indian Kanoon. The platform runs on classical term-matching, and the workhorse behind it is BM25 [8], a probabilistic ranking function that scores documents by term frequency scaled against inverse document frequency across the corpus, and that we use here as the standard sparse baseline against which neural alternatives are measured. Simple as the formulation sounds, BM25 is stubbornly hard to beat. Empirical comparisons on legal retrieval benchmarks show it matching or outperforming several neural alternatives, particularly when queries contain distinctive legal phrases. We suspect the community often reaches for neural approaches prematurely, without first exhausting what a well-tuned BM25 can deliver. The failure mode, however, is precise and consequential, namely vocabulary mismatch. A query about “breach of fiduciary duty” returns nothing when the relevant judgment phrases the same concept as “violation of trust obligations,” and no amount of IDF reweighting can bridge a gap where the surface terms simply do not overlap — though, frankly, where exactly the mismatch tips from inconvenient to catastrophic is a line we have not seen the literature pin down with any precision.

B. Dense Retrieval

Dense retrieval attacks the vocabulary mismatch problem head-on. Queries and documents are encoded into continuous vector representations, and similarity is computed via cosine or dot-product in the embedding space. Transformer-based encoders learn to push semantically related texts close together, so “breach of fiduciary duty” and “violation of trust obligations” end up in the same neighbourhood despite sharing no surface tokens. Dual-encoder architectures with summarisation-based re-ranking have proven competitive on the COLIEE case law retrieval task [9]. Scaling this to real corpora, however, is a separate problem. FAISS [10], an approximate nearest-neighbour library that supports sub-linear search over billion-scale vector collections on commodity GPUs, supplies the standard infrastructure for dense retrieval at production scale. The cost is non-trivial: index construction and query-time inference both demand significantly more compute than BM25’s lightweight inverted indices, and we are honestly not yet sure whether this trade-off pays off for Indian legal corpora, which are large but perhaps not yet at the scale where sparse retrieval visibly collapses.

C. Domain-Adapted Models for Indian Legal Retrieval

A BERT checkpoint trained on Wikipedia and BooksCorpus knows very little about *ratio decidendi* or “Section 302

IPC.” Legal text is unusual enough — long sentences, nested citations, Latin-inflected phrasing — that general-purpose encoders consistently underperform on legal retrieval and classification tasks. The fix is straightforward in principle, namely continued pre-training on in-domain data. InLegalBERT [5], a BERT-based encoder pre-trained on a large corpus of Supreme Court and High Court judgments, does exactly this and is used as the domain-adapted backbone in most subsequent Indian legal retrieval and classification work. It consistently outperforms both vanilla BERT and multilingual variants across Indian legal benchmarks. We read this result as confirmation that domain shift in legal NLP is not a minor nuisance; it is a first-order problem that general-purpose scale alone does not solve. Whether the right framing here is “BERT was always going to fail on Indian law” or “BERT could have been salvaged with enough adapter training, and we never really tested that alternative at scale,” we are not, in candour, fully sure.

D. Retrieval-Augmented Generation for Legal Reasoning

RAG has become the dominant pattern for grounding language model outputs in retrieved evidence, and legal reasoning is a natural fit for it. Instead of relying on whatever a large language model happens to have memorised during pre-training, the generative component receives actual precedents as context. NyayaRAG [1], a retrieval-augmented framework built specifically over Indian Supreme Court and High Court corpora, applies this directly to judgment prediction and shows that retrieval-grounded generation substantially outperforms standalone generation across multiple court levels. CBR-RAG [3], which integrates case-based reasoning with retrieval-augmented generation, goes a step further: rather than retrieving cases that are merely semantically close, it surfaces structurally analogous cases, improving the legal relevance of the context window. These are meaningful advances. However, we note a persistent structural limitation: in every existing Indian legal AI system we have surveyed, retrieved outputs feed into a generation module and stop there. They are never routed onward to structured fact analysis, argument construction, or outcome prediction. Retrieval remains an endpoint rather than an input to a broader reasoning chain — and we have spent some time trying to articulate why this is, without arriving at an answer that fully satisfies us.

IV. CASE FACT STRUCTURING

An Indian Supreme Court judgment can run to tens of thousands of words. Procedural history, factual narrative, statutory analysis, counsels’ submissions, and the final order are all woven together without consistent formatting or explicit section markers. Feed that to a prediction model as raw text, and you are asking it to find a needle in a haystack while blindfolded. We argue that structuring, that is, the decomposition of the document into its functional parts, is a prerequisite rather than a convenience. Three structuring tasks have received sustained attention in Indian legal NLP,

namely named entity recognition, rhetorical role classification, and statutory section tagging.

A. Legal Named Entity Recognition

General-purpose NER (person, location, organisation) is almost useless on a court judgment. The entities that actually matter here are judge names, petitioner and respondent roles, case numbers, statute references, and cited precedents. Kalamkar et al. [11] recognised this and built a fourteen-type schema designed specifically for Indian judicial text, covering courts, judges, petitioners, respondents, lawyers, dates, case numbers, statutes, provisions, precedents, witnesses, other persons, organisations, and geographical entities. Transformer-based sequence labelling models were evaluated against this benchmark on Supreme Court and High Court judgments. The downstream payoff is immediate. Precedent entities can be linked to their source judgments for retrieval, party entities feed coreference resolution, and statute entities bridge to the statutory knowledge base. Without this scaffolding, every downstream module is forced to operate over raw token sequences with no structured hooks to latch onto. We find it somewhat surprising that legal NER still receives relatively little attention compared to classification tasks, given how much of the pipeline depends on it — although, on reflection, perhaps the issue is that NER feels solved when in fact it is only solved on the first three or four entity types, and the rest are quietly carrying error that nobody, ourselves included, has audited.

B. Rhetorical Role Classification

Every sentence in a judgment plays a role: stating facts, citing a prior ruling, framing a legal issue, articulating reasoning, or delivering the operative decision. Rhetorical role labelling assigns these functional categories automatically. LegalEval (SemEval-2023 Task 6) [4] established a multi-court benchmark for this task across Indian jurisdictions, and the results were instructive: automated segmentation is feasible, though far from solved. The hard part is context dependence. “The appeal is dismissed” means one thing when it appears inside a recitation of a prior ruling and something entirely different as the operative order of the present case. You cannot classify the sentence in isolation; long-range context is required. Hierarchical and sequential models that propagate information across sentence boundaries help, but we observe that most current approaches still struggle at document lengths typical of Indian high court judgments. Rhetorical segmentation matters because it is essentially the only way to isolate the *ratio decidendi* from the surrounding noise; without it, every sentence carries equal weight, which we keep flagging as the obvious objection but have not yet seen any system in this space fully address.

C. IPC Section Tagging and Statute Linking

Statute identification asks a deceptively simple question: which sections of the Indian Penal Code (or other legislative instruments) does this judgment actually engage with?

LegalEval [4] includes statute identification as one of three subtasks, with annotated data and evaluation protocols. The task matters because it is the bridge between facts and law; without knowing which statutes apply, neither argument construction nor outcome prediction can proceed on solid footing. The persistent difficulty is implicit reference. Judges routinely discuss the elements, interpretation, and applicability of a statute without ever writing down the section number, leaving the model to perform semantic matching between case language and statutory text — which is considerably harder than spotting an explicit “Section 302” mention, and where, candidly, the line falls between “the model has correctly inferred the statute” and “the model has guessed plausibly” is a question we have not yet seen anyone evaluate carefully.

Taken together, NER, rhetorical roles, and statute tagging address complementary dimensions of document structure: who, what role, and under what law. All three should feed into downstream reasoning as intermediate representations. In practice, they do not. Each task has been studied in isolation within Indian legal NLP, and no existing system propagates their combined outputs into a unified pipeline. We are left with a structuring layer that produces terminal outputs, carefully extracted, and then abandoned.

V. ARGUMENT GRAPH CONSTRUCTION

This is the axis where the literature thins out sharply. Legal reasoning is, at its core, argumentative. A decision crystallises from competing claims, supporting evidence, statutory authority, and precedent invoked by analogy or distinction. If you can represent these components, and the relations between them — support, attack, citation, analogy — as a directed graph, then you have made the inferential skeleton of a case explicit and machine-readable. We consider argument graph construction the most structurally ambitious of the five axes in this survey. It is also, by a wide margin, the one with the least Indian legal AI work behind it. What follows therefore draws heavily on the broader argument mining literature and on empirical work conducted in Western jurisdictions.

A. Argument Mining: Foundations

Argument mining extracts claims, premises, and evidence from text, and maps the relations between them: support, attack, rebuttal. The most comprehensive legal application to date is Habernal et al.’s work on European Court of Human Rights decisions [12], in which a taxonomy of seventeen argumentative component types was used to annotate and train transformer-based extractors. Performance is reasonable on shorter decisions but degrades noticeably as documents grow longer and argumentative structure becomes more tangled. The field has moved from sentence-level classification (is this sentence a claim or a premise?) to graph-level modelling, where nodes represent argumentative units and directed edges encode support, contradiction, and evidential relations. This shift matters enormously for legal applications. A single precedent might support a petitioner’s claim while simultaneously undermining the respondent’s, and only a graph can capture

that kind of ambivalence — though we should admit we have not actually verified, on Indian data, that ECHR-trained extractors transfer at all, and the longer that omission sits in our own diagnosis, the more it begins to feel like a hole in it.

B. Argument Structure in Legal Corpora

Almost all empirical work on legal argument mining has been conducted on European and American case law. LexGLUE [13], a benchmark suite for legal language understanding that is now the closest thing the field has to a standard, evaluates primarily on ECHR and other Western corpora. Indian courts are simply not represented. That is not a minor gap; it is a blind spot. The structure of a legal argument graph maps naturally onto the components of an Indian judgment: claims correspond to legal issues in dispute, evidence nodes capture trial-established facts, statute nodes provide the normative warrant, and precedent nodes supply analogical support or distinguishing authority. With a complete graph in hand, weak-point detection — the task of identifying claims that lack adequate evidentiary or statutory backing — becomes a matter of graph completeness analysis. We think this capability alone would be worth the annotation investment, even if no further downstream use were envisioned, although we acknowledge we have been making this pitch for some time without anyone funding the annotation, and at some point that pattern itself becomes data we should probably take seriously.

C. The Indian Legal Gap and Graph Database Representations

As of this writing, no published dataset, annotation schema, or trained model addresses argument graph construction for Indian legal judgments. The gap is total. Several factors explain it. Indian judgments are substantially longer and more digressive than the ECHR decisions that existing argument mining models were designed for. Their argumentative structure is less formulaic; reasoning is often scattered across non-contiguous sections, which makes span extraction unreliable. The most fundamental barrier, however, is the simplest: annotated training data for Indian courts does not exist.

On the representational side, graph databases such as Neo4j are a natural fit: nodes for claims, evidence, statutes, and precedents, with typed edges for support, citation, contradiction, and analogy. The architecture, in some sense, almost designs itself. What makes this axis particularly compelling from a systems perspective is that its inputs already exist, in principle. Retrieved precedents from Section III, NER entities, rhetorical segments, and statute tags from Section IV all feed naturally into argument graph construction. This axis is the connective tissue of the pipeline. Without it, the outputs of retrieval and structuring have nowhere coherent to go. We consider Indian legal argument mining a high-priority research direction, and we are frankly surprised it has received so little attention — though, having sat with the surprise long enough, we are starting to think “surprise” is no longer the right word, and that perhaps the more concerning state of affairs is that we have stopped being surprised at all.

VI. JUDICIAL OUTCOME SIMULATION

Prediction has the longest research history of any axis in this survey, moving steadily from feature-engineered classifiers through transformer-based encoders to the current crop of legal large language models. The task itself is straightforward to state: take the text of a case, predict whether the appeal was accepted or dismissed, and where possible, generate the reasoning behind that prediction. We are, we believe, at a transition point where the community has begun to recognise that accuracy numbers alone, taken in isolation, mean very little in a legal context. A model that predicts the right outcome for the wrong legal reasons is not a tool; it is a liability.

A. Classical and Early Neural Approaches

The early period of legal judgment prediction was defined by manual feature engineering. Researchers spent months hand-crafting indicators for bag-of-words and TF-IDF models, and the resulting systems worked, after a fashion. They were also brittle, failing the moment they encountered a case from a different court or a different decade. Neural networks, specifically CNNs and RNNs, removed the need for manual features but ran into a different wall, namely document length. Indian judgments are long. Most early recurrent models either ran out of memory or quietly lost the relevant context by the time they reached the operative order. LegalBERT was the first real breakthrough on this front, demonstrating that domain shift in legal text is severe enough that general-purpose embeddings effectively encode a different language.

B. Transformer-Based Prediction

Transformers changed what was possible. Through self-attention, models such as InLegalBERT [5], a BERT-based encoder pre-trained on Indian Supreme Court and High Court judgments, could finally capture dependencies between a factual assertion on page two and a statutory application on page twenty. Foundational benchmarking work by [6] established the ILDC corpus, a collection of over thirty-five thousand Supreme Court cases annotated for binary judgment prediction, and made systematic comparison across architectures possible for the first time. The 512-token limit of base BERT, however, remains a persistent frustration in this line of work. We have seen researchers try almost everything (overlapping chunks, sliding windows, hierarchical pooling) but we suspect this ceiling is more damaging than current evaluations reveal — you cannot, in any honest sense, model a hundred-page high court judgment by looking at it through a series of disconnected five-hundred-word keyholes.

C. Explainability as a First-Class Requirement

A model that outputs “dismissed” without an accompanying reason is, for legal use, of little value. You cannot audit it, you cannot identify its biases, and you certainly cannot build trust with it. PredEx [7], the largest expert-annotated corpus for legal judgment prediction and explanation in the Indian context, addressed this directly by providing roughly

fifteen thousand cases in which the rationales are not model-generated hallucinations but actual explanations written by legal experts. This shifts the evaluation question from “does the output look plausible?” to “is the output grounded in expert reasoning?” We observe, however, that even these expert annotations are often compressed summaries of a much longer reasoning process, and whether a summary can really capture the recursive, self-correcting character of a judge’s internal deliberation is, in candour, something we have started to wonder about more often than we used to.

D. Large Language Models and the Scale Frontier

Scale has become the default answer to almost everything in NLP, and legal AI is no exception. *NyayaAnumana*, together with its specialised companion model *INLegalLlama* [1], the largest and most diverse Indian legal corpus compiled to date, currently represents the ceiling for Indian legal judgment prediction, with the language model internalising the rhythms of judicial writing across roughly seven hundred thousand case proceedings spanning multiple court levels. Scale, however, has brought a new problem along with it: benchmark bias. Recent work has shown that many of our current test sets are drawn disproportionately from cases with unambiguous outcomes, which makes models look considerably better in evaluation than they appear to be in the messier reality of a contested courtroom [14]. We believe benchmark bias is the single most underappreciated methodological issue in the field today.

E. Remaining Limitations

Three gaps persist. First, prediction models are still built in isolation from the retrieval and structuring axes discussed in earlier sections. There is no established pathway for a structured fact representation, or a retrieved precedent, to inform a final prediction in any principled way.

Second, the available data is heavily Supreme Court weighted. District courts are where the real backlog lives, yet they have almost no presence in our current benchmarks.

Third, the evaluation metrics are often lazy. Accuracy is the easiest thing to measure, but in a legal context, it is also the most misleading. A model that achieves ninety percent accuracy but fails systematically every time a witness is deemed unreliable is not merely imperfect; it is dangerous, precisely because nobody has thought to check that specific failure mode.

VII. WHAT-IF ANALYSIS

What-if analysis is the capability that practitioners most often ask for, and it is also the axis with almost no published literature behind it. It is, in our view, the axis that matters most. Where prediction tells you what the likely outcome is, and explainability tells you why, what-if analysis asks what the outcome would have been had the facts, the statutes invoked, or the precedents relied upon been different. For a lawyer preparing a case, counterfactual reasoning is the form of explainability that actually matters.

A. Counterfactual NLP

In general NLP, counterfactual analysis usually means perturbing an input to see when the predicted label flips. If a “dowry demand” is recharacterised as a “loan request,” does the conviction still stand? This is more than a technical curiosity. It is a way to map the decision boundaries of a model and to identify which facts are actually driving the predicted outcome. We distinguish, loosely, between forward-looking what-if analysis (testing hypotheses about how a case might be argued) and backward-looking counterfactuals (explaining past decisions in terms of the minimal changes that would have flipped them).

B. XAI in Legal Decision-Making

The legal system imposes a transparency requirement that most AI systems are not built to meet. A litigant has, in principle, the right to know why an interest was affected, especially when an algorithm contributed to that decision [15]. Most current XAI methods, including LIME and SHAP, are designed to surface feature importance, not the structured counterfactuals that legal reasoning actually requires. Until systems can answer “what if” rather than only “why,” their explanations will remain descriptive rather than prescriptive.

C. The Indian Gap

There is no published what-if analysis system for Indian law. We searched, and we found nothing.

This is a substantial opportunity for the community, particularly given the resource constraints typical of Indian litigation, where a tool that lets a lawyer test alternative statutory framings or fact patterns without commissioning a new dataset each time would be genuinely transformative. It is also, we should say, the logical endpoint of the integration we have been arguing for throughout this survey.

Our survey ends here because the literature ends here. We cannot review what has not been built. We can, however, point to the empty space and call it a priority.

VIII. DATASETS

Datasets have been the bottleneck. Western legal AI inherited decades of structured case law from repositories such as LexisNexis and PACER; Indian legal NLP has had to build its benchmarks from scratch, one annotation campaign at a time, typically depending on legal professionals whose time is expensive and scarce. We survey the seven principal datasets available as of 2025. They differ sharply in scale, court coverage, annotation depth, and public availability, and we believe these differences matter more than they are usually credited with: where the field claims to stand and where it actually stands diverge along precisely these axes.

The jump from ILDC’s thirty-five thousand Supreme Court cases, sourced from Indian Kanoon and annotated with binary outcome labels alongside multi-sentence explanations [6], to *NyayaAnumana*’s seven hundred thousand judgments, drawn from the Supreme Court, the High Courts, and the District Courts and shipped with outcome labels plus court-level

TABLE II
PRIMARY DATASETS FOR INDIAN LEGAL NLP RESEARCH (2021–2025)

Dataset	Reference	Size	Court Coverage	Annotation Type	Avail.
ILDC	[6]	35,000 cases	Supreme Court of India	Binary outcome labels, multi-sentence explanations	Public
PredEx	[7]	15,000+ cases	Supreme Court of India	Expert-annotated natural language explanations aligned to outcomes	Public
NyayaAnumana	[1]	700,000+ cases	Supreme Court, High Courts, District Courts	Outcome labels, court-level metadata	Public
NyayaRAG Corpus	[2]	Subset of Indian judgments	Supreme Court and High Courts	Retrieval-annotated with relevant precedent links	Partial
LegalEval (SemEval-2023 Task 6)	[4]	9,380 sent. (NER); 14,864 sent. (roles)	Supreme Court and High Courts	NER labels (14 types), rhetorical role labels, statute tags	Public
InLegalBERT Corpus	[5]	5.4M sentences	Supreme Court and High Courts	Pre-training corpus, no task-specific labels	Partial
Indian Kanoon	–	Millions of documents	All court levels	No annotation, raw text only	Public

metadata [1], is not incremental. It is a qualitative shift that, for the first time, makes LLM-scale fine-tuning over Indian legal text genuinely feasible. Scale and depth, however, are not the same thing. NyayaAnumana provides outcome labels and court-level metadata at volume; it does not include the expert-written natural language explanations that PredEx [7], a fifteen-thousand-case Supreme Court corpus annotated by legal experts with explanations aligned to outcomes, supplies at a fraction of the scale. We would argue that any claim about “state-of-the-art” Indian legal judgment prediction needs to specify whether the model was trained with explanation supervision or without it, because the two settings produce fundamentally different systems and the literature has been sloppy about treating them as comparable.

District court data, meanwhile, is almost invisible across all annotated datasets, a glaring omission given that district courts handle the vast majority of India’s pending caseload. The InLegalBERT pre-training corpus [5], roughly 5.4 million sentences drawn from Supreme Court and High Court judgments and shipped without task-specific labels, together with the broader Indian Kanoon collection of millions of unannotated documents, supplies raw text at scale. With no task-specific annotations, however, their utility is limited to pre-training and unsupervised exploration.

Three gaps stand out, and none of the seven datasets we surveyed addresses any of them. First, there are no argument-level annotations linking claims to evidence, statutes, and precedents within a single judgment, which forces argument graph construction (Section V) to operate in an unsupervised or weakly supervised mode. Second, there are no multilingual annotations that handle the Hindi–English or regional-language code-mixing pervasive in Indian judicial text, despite this code-mixing being routine rather than rare. Third, there are no counterfactual annotations of the kind a what-if analysis

system (Section VII) would need for either training or evaluation. Each of these gaps maps directly onto an open research direction we have flagged earlier in this paper. Closing them will require annotation campaigns that bring NLP researchers and legal domain experts together at a scale the community has not yet managed, and we are honestly not sure what the right organisational form for that collaboration looks like [4].

IX. EVALUATION METRICS

How a system is measured matters at least as much as how it is built, and this is a point that legal AI papers tend to gloss over. Pick the wrong metric and you end up optimising for something the legal system does not actually care about, or worse, producing a model that posts impressive leaderboard numbers while failing on precisely the cases that matter most. The majority of legal AI studies report accuracy as their sole performance measure, without justifying that choice or acknowledging what it conceals [16]. On imbalanced datasets, and Indian legal datasets are almost always imbalanced, accuracy is close to meaningless. We survey the metrics used across the retrieval, classification, and generation tasks covered in this paper, and where appropriate, we are not shy about flagging where metric choice has led the field astray.

A. Retrieval Metrics

MAP (Mean Average Precision) remains the default for ranked retrieval evaluation. It computes precision at each relevant document in the ranked list, averages across the list, and then averages again across queries, yielding a single number that captures both how many relevant documents were retrieved and how high they were ranked. nDCG (Normalised Discounted Cumulative Gain) complements MAP by applying a logarithmic discount to lower-ranked results, which aligns reasonably well with how lawyers actually search: they look at the first page, perhaps the second, and rarely venture further.

Both BM25 evaluations [8] and dense retrieval experiments in COLIEE [9] report MAP and nDCG as primary measures.

The catch, and we believe this is underappreciated, is that both metrics require relevance judgments. In legal retrieval, relevance is not a surface-level property. Whether a 1987 Supreme Court ruling on tenant eviction is “relevant” to a 2022 high court dispute over landlord obligations is a question of legal interpretation, not of keyword overlap. Collecting those judgments at scale requires legal expertise, which means that evaluation itself is bottlenecked by exactly the same annotation scarcity that constrains training data, and we have not yet seen a methodology that genuinely works around this rather than around the edges of it.

B. Classification, Generation, and Explainability Metrics

Accuracy and macro-averaged F1 are the standard pair for binary and multi-class outcome prediction. On balanced data, accuracy is acceptable. Indian legal datasets, however, are rarely balanced. Dismissals dominate most corpora, which means a classifier that predicts “dismissed” every single time can post a respectable accuracy score without having learned anything at all [16]. Macro-F1 corrects for this by weighting each class equally, and we would argue, plainly, that it should be the primary metric in every legal judgment prediction paper, not a secondary number buried in an appendix.

NER evaluation requires a finer granularity than aggregate F1 can provide. An aggregate score across all entity types can hide the fact that a model performs well on common categories such as dates and court names while failing almost completely on rare but legally critical ones such as witness names and provision numbers [11]. Per-entity-type F1 is, in our view, the minimum acceptable standard; anything less is misleading.

Explanation evaluation is harder, because the property being measured, namely legal soundness, is not something an automated metric can assess. ROUGE and BERTScore capture surface overlap with reference text, which tells you whether the generated explanation reuses similar words. They tell you nothing about whether the underlying reasoning is correct. Expert Likert-scale assessments, in which legal professionals rate adequacy, accuracy, and completeness on structured scales, remain the only reliable approach. Both INLegalLLama [1] and PredEx [7] adopted this methodology. It is expensive. It does not scale well. And it is, as far as we can tell, irreplaceable, which is a problem the community has not yet confronted honestly.

X. GAPS AND CONTRIBUTION

We can state the central finding of this survey plainly: nobody has built it yet. No existing Indian legal AI system connects precedent retrieval to case fact structuring, structuring to argument graph construction, argument graphs to outcome prediction, or prediction to counterfactual what-if analysis. Each axis we surveyed operates within its own silo. Retrieved precedents are never routed to a structuring module. Structured facts and extracted entities stop at the output layer; no argument graph constructor picks them up. Prediction models

run without argument-level context, and their outputs are never fed into any counterfactual reasoning component. The closest the field has come to integration is coupling retrieval with generation [2] or with case-based reasoning [3], but even those systems span only two of the five axes. We have come to think this fragmentation is no longer a reflection of the field’s youth; it is a design choice the community has been making by default, and one that needs to be made deliberately instead.

At the task level, three gaps stand out. Argument graph construction for Indian legal judgments has no published dataset, no trained model, and no evaluation benchmark. It is, by any measure, the largest structural blind spot in the field. Counterfactual what-if analysis has not been attempted at all in the Indian context, even though it is arguably the capability most directly relevant to practising lawyers, the people these systems are nominally being built for. And evaluation remains systematically tilted toward Supreme Court data [6], [7]. District and high courts, which together account for the overwhelming bulk of India’s pending caseload, lack reliable benchmarks. LegalEval’s annotation scope reflects the same imbalance [4].

The data-level picture is no better. No annotated resource captures argument-level structure, that is, the claim-evidence-statute-precedent linkages that argument graph construction would require. Multilingual annotations are absent entirely; Hindi and regional language passages embedded in English judgments remain invisible to every model we surveyed. The large-scale corpora that enable LLM fine-tuning, with NyayaAnumana chief among them [1], trade annotation depth for volume. PredEx, by contrast, offers expert-level explanations but at a fraction of the scale [7]. No single resource delivers both. We suspect that until one does, the field will continue producing systems that are either large-scale but shallow, or deep but narrow.

We should be clear about what this paper contributes and what it does not. It is a survey, not a system. We offer the first structured review of Indian legal NLP organised around five functional capability axes, map the existing literature to each axis, catalogue the datasets and evaluation methods currently in use, and identify what remains unsolved. Our primary claim is that individual capabilities have matured enough to make integration feasible, and that the absence of integration, rather than the immaturity of any one axis, is now the binding constraint on meaningful progress. We hope this survey serves as a useful reference for anyone aiming to build toward a unified Indian legal AI architecture. We make no claim that building one will be easy.

XI. ETHICAL CONSIDERATIONS

Train a model on historical Indian court judgments and it will learn what those judgments contain, including their biases. Outcome disparities correlated with gender, caste, economic status, and geography are well documented in Indian legal history, and a prediction system optimised to reproduce historical outcome distributions will reproduce historical inequities along precisely these lines, unless someone intervenes

deliberately during training and evaluation. The problem is not unique to India; bias amplification has been flagged across jurisdictions [16]. The stakes, however, escalate sharply once model outputs begin to influence real legal decisions, because there is a genuine risk that statistical artefacts get mistaken for legal reasoning by users who trust the system more than they should. We believe bias auditing across demographic and geographic subgroups should be a non-negotiable prerequisite for deployment, not a post-launch afterthought, and we say this with some conviction because the alternative, treating it as something to “come back to,” is a failure mode we have already seen play out elsewhere.

Explainability, in a legal setting, is not a feature. It is a procedural obligation. Parties to litigation have the right to understand the basis of decisions that affect their legal interests, and that right does not evaporate the moment part of the decision process becomes automated [15]. This creates a genuine difficulty for the field, because the models that predict best are often the ones that explain least. A generated paragraph that reads fluently and cites the right statutes may still be legally incoherent, and no automated metric will catch that. PredEx’s methodology, grounding explanation evaluation in expert annotation rather than ROUGE scores [7], points in the right direction. But the community still needs to internalise a harder lesson, namely that fluency is not soundness. A model that writes well but reasons poorly is more dangerous, not less, because its confidence is convincing.

One distinction must be preserved. Everything we have surveyed in this paper, including retrieval systems, structuring modules, prediction models, and explanation generators, belongs squarely in the category of decision-support tools. They organise information, surface relevant precedents, structure arguments, and estimate outcomes so that human legal professionals can reason more effectively. They are not, and they should not be positioned as, autonomous adjudicators. The most capable systems currently available still exhibit significant limitations in coverage, interpretability, and generalisation that rule out unsupervised deployment [1], [17]. Building Indian legal AI responsibly will require sustained collaboration among NLP researchers, legal professionals, and policymakers. The goal is augmentation, not automation, and we believe losing sight of that distinction is the fastest way to erode the procedural safeguards on which the rule of law depends.

REFERENCES

- [1] S. K. Nigam, D. P. Balaramamahanthi, S. Mishra, N. Shallum, K. Ghosh, and A. Bhattacharya, “NYAYAANUMANA and INLEGALLAMA: The largest Indian legal judgment prediction dataset and specialized language model for enhanced decision analysis,” in *Proceedings of the 31st International Conference on Computational Linguistics*. Abu Dhabi, UAE: Association for Computational Linguistics, 2025, pp. 11 135–11 160. [Online]. Available: <https://aclanthology.org/2025.coling-main.738/>
- [2] S. K. Nigam, B. D. Patnaik, S. Mishra, A. V. Thomas, N. Shallum, K. Ghosh, and A. Bhattacharya, “NyayaRAG: Realistic legal judgment prediction with RAG under the Indian common law system,” in *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*. Mumbai, India: Association for Computational Linguistics, 2025, pp. 1709–1726. [Online]. Available: <https://aclanthology.org/2025.ijcnlp-long.92/>
- [3] N. Wiratunga, R. Abeyratne, L. Jayawardena, K. Martin, S. Massie, I. Nkisi-Orji, R. Weerasinghe, A. Liret, and B. Fleisch, “CBR-RAG: Case-based reasoning for retrieval augmented generation in LLMs for legal question answering,” in *Case-Based Reasoning Research and Development. ICCBR 2024*, ser. Lecture Notes in Computer Science, vol. 14775. Springer, Cham, 2024, pp. 445–460.
- [4] A. Modi, P. Kalamkar, S. Karn, A. Tiwari, A. Joshi, S. K. Tanikella, S. K. Guha, S. Malhan, and V. Raghavan, “SemEval-2023 task 6: LegalEval – understanding legal texts,” in *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*. Toronto, Canada: Association for Computational Linguistics, 2023, pp. 2362–2374. [Online]. Available: <https://aclanthology.org/2023.semeval-1.318/>
- [5] S. Paul, A. Mandal, P. Goyal, and S. Ghosh, “Pre-trained language models for the legal domain: A case study on Indian law,” in *Proceedings of the 19th International Conference on Artificial Intelligence and Law*. ACM, 2023, pp. 187–196. [Online]. Available: <https://arxiv.org/abs/2209.06049>
- [6] V. Malik, R. Sanjay, S. K. Nigam, K. Ghosh, S. K. Guha, A. Bhattacharya, and A. Modi, “ILDC for CJPE: Indian legal documents corpus for court judgment prediction and explanation,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Online: Association for Computational Linguistics, 2021, pp. 4046–4062. [Online]. Available: <https://aclanthology.org/2021.acl-long.313/>
- [7] S. K. Nigam, A. Sharma, D. Khanna, N. Shallum, K. Ghosh, and A. Bhattacharya, “Legal judgment reimaged: PredEx and the rise of intelligent AI interpretation in Indian courts,” in *Findings of the Association for Computational Linguistics: ACL 2024*. Bangkok, Thailand: Association for Computational Linguistics, 2024, pp. 4296–4315. [Online]. Available: <https://aclanthology.org/2024.findings-acl.255/>
- [8] G. M. Rosa, R. C. Rodrigues, R. Lotufo, and R. Nogueira, “Yes, BM25 is a strong baseline for legal case retrieval,” *arXiv preprint arXiv:2105.05686*, 2021. [Online]. Available: <https://arxiv.org/abs/2105.05686>
- [9] S. Althammer, A. Askari, S. Verberne, and A. Hanbury, “DoSSIER@COLIEE 2021: Leveraging dense retrieval and summarization-based re-ranking for case law retrieval,” *arXiv preprint arXiv:2108.03937*, 2021. [Online]. Available: <https://arxiv.org/abs/2108.03937>
- [10] J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [11] P. Kalamkar, A. Agarwal, A. Tiwari, S. Gupta, S. Karn, and V. Raghavan, “Named entity recognition in Indian court judgments,” in *Proceedings of the Natural Legal Language Processing Workshop 2022*. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, 2022, pp. 184–193. [Online]. Available: <https://aclanthology.org/2022.nllp-1.15>
- [12] I. Habernal, D. Faber, N. Recchia, S. Weiss, I. Gurevych, C. Vogler, and L. Rothenberger, “Mining legal arguments in court decisions,” *Artificial Intelligence and Law*, 2023. [Online]. Available: <https://arxiv.org/abs/2208.06178>
- [13] I. Chalkidis, A. Jana, D. Hartung, M. Bommarito, I. Androustopoulos, D. M. Katz, and N. Aletras, “LexGLUE: A benchmark dataset for legal language understanding in English,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Dublin, Ireland: Association for Computational Linguistics, 2022, pp. 4317–4335. [Online]. Available: <https://aclanthology.org/2022.acl-long.297/>
- [14] S. K. Nigam, B. D. Patnaik, S. Mishra, N. Shallum, K. Ghosh, and A. Bhattacharya, “Rethinking legal judgement prediction in a realistic scenario in the era of large language models,” *arXiv preprint arXiv:2410.10542*, 2024. [Online]. Available: <https://arxiv.org/abs/2410.10542>
- [15] S. Deakin and S. Markou, “Explainable AI and law: An evidential survey,” *Digital Society*, vol. 2, no. 2, p. 27, 2023.
- [16] M. Medvedeva, M. Vols, and M. Wieling, “Using machine learning to predict decisions of the European court of human rights,” *Artificial Intelligence and Law*, vol. 28, no. 2, pp. 237–266, 2020.

- [17] M. Mersha, S. Sprunger, T. R. Hoens, R. Nwali, D. Aidoo, T. Pham, and T. Warnow, "Explainable artificial intelligence: A survey of needs, techniques, applications, and future direction," *arXiv preprint arXiv:2409.00265*, 2024. [Online]. Available: <https://arxiv.org/abs/2409.00265>